TU Dortmund University

Department of Statistics

Gender and Parental Education Effects on Student Test Performance: A Statistical Analysis

Application Report for the
Master of Science in Data Science

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1 Introduction

Do boys and girls perform differently in Mathematics and Language? And does it matter whether a student's parents went to university or stopped after high school? These are not new questions — PISA data has shown for years that boys tend to outscore girls in mathematics while girls do better in reading (OECD, 2023), and a large meta-analysis by Sirin (2005) confirms that parental education is one of the strongest predictors of academic achievement. But these patterns vary by context, and it is worth checking whether they hold in a specific, smaller dataset before assuming they do.

This report examines both questions using data on 486 students who were tested in Mathematics and Language. Two research questions guide the analysis:

RQ1. Is there a statistically significant difference between male and female students in their Mathematics scores and in their Language scores?

RQ2. Is there a statistically significant difference in test scores across parental education levels? If so, which specific groups differ from one another?

In short, the answer to both is yes. Males score higher in Mathematics, females score higher in Language, and the biggest gap in parental education is between the high school group and everyone else — not between higher levels of education. The rest of the report is organised as follows: Section 3 defines the statistical methods, Section 4 presents the results, and Section 5 discusses what these findings mean.

2 Problem Description

2.1 Data Source and Collection

The dataset used in this analysis was provided by the Department of Statistics at TU Dortmund University for the Winter Semester 2026/2027 application process. It is available in the file `Scores.csv` and contains records for 486 students. The data are structured in long format, with each student contributing two rows — one for their Mathematics score and one for their Language score — yielding a total of 972 observations. For the purposes of analysis, the dataset was reshaped into wide format so that each student is represented by a single row containing both scores. All statistical analyses were conducted in Python 3 (Van Rossum & Drake, 2009) using the `pandas` (McKinney, 2010), `NumPy` (Harris et al., 2020), `SciPy` (Virtanen et al., 2020), `Matplotlib` (Hunter, 2007), and `seaborn` (Waskom, 2021) libraries. The complete analysis code is available at: `https://github.com/hirarehman161/Gender-and-Parental-Education-Effects-on-Student-Test-Performance`.

2.2 Variables

The dataset contains the following variables:

- **gender**: A binary categorical variable taking values *female* (n = 245) and *male* (n = 241).
- **parental.level.of.education**: An ordinal categorical variable with four levels, ordered from lowest to highest qualification: *high school* (n = 156), *associate's degree* (n = 177), *bachelor's degree* (n = 94), and *master's degree* (n = 59).
- **math_score**: A continuous variable representing the student's Mathematics test score, measured on a scale from 0 to 100.
- **language_score**: A continuous variable representing the student's Language test score, also measured on a scale from 0 to 100.

No missing values are present in any variable. The two score variables are ratio-scaled, making them suitable for both parametric and non-parametric statistical procedures.

2.3 Research Questions

RQ1 involves a two-group comparison (female vs. male), addressed by a two-sample test of means. RQ2 involves four groups defined by parental education level, requiring an omnibus test followed by pairwise post-hoc comparisons to identify which specific groups differ.

3 Methods

3.1 Descriptive Statistics

Let $x_1, x_2, \dots, x_n$ denote a sample of n observed scores. The following measures are used to characterise the distribution of each group.

The **sample mean** provides a measure of central tendency:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{1}$$

The **sample variance** and its square root, the **sample standard deviation** s, measure the spread of the data around the mean:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \tag{2}$$

The **median** $\tilde{x}$ is the value that divides the ordered sample into two equal halves, and is a robust measure of central tendency when the distribution is skewed. The **interquartile range** (IQR) is defined as $\text{IQR} = Q_3 - Q_1$, where Q_1 and Q_3 are the first and third quartiles, respectively (Montgomery & Runger, 2014).

Boxplots are used to visualise the distribution of scores across groups. Each boxplot displays the median as a horizontal line within the box, the box boundaries representing Q_1 and Q_3, and whiskers extending to the most extreme values within $1.5 \times \text{IQR}$ of the quartiles. Observations beyond this range are plotted individually as outliers (Tukey, 1977).

3.2 Assumption Checks: Shapiro-Wilk and Levene's Tests

Before selecting hypothesis tests, normality is assessed via the Shapiro-Wilk test (Shapiro & Wilk, 1965). Given order statistics $x_{(1)} \leq \cdots \leq x_{(n)}$, the statistic is:

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \tag{3}$$

where a_i are coefficients derived from the expected values and covariance matrix of standard normal order statistics. $W \in (0, 1]$; values near 1 support normality. A p-value below $\alpha = 0.05$ rejects normality. The test is valid for $n \leq 2000$ (Royston, 1992). Homogeneity of variance is assessed via Levene's test (Levene, 1960), which tests $H_0 : \sigma_1^2 = \cdots = \sigma_k^2$ by applying an F-test to the mean absolute deviations from group means. A p-value above 0.05 confirms the equal-variance assumption.

3.3 Two-Sample *t*-Test for Gender Comparisons (RQ1)

When both groups are approximately normally distributed and have equal variances, the appropriate test for comparing two independent group means is the two-sample Student's t-test (Student, 1908). Let group 1 (female) have sample size n_1, mean $\bar{x}_1$, and standard deviation s_1, and group 2 (male) have corresponding values n_2, $\bar{x}_2$, and s_2.

The pooled standard deviation is:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \tag{4}$$

The test statistic is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \tag{5}$$

Under the null hypothesis $H_0 : \mu_1 = \mu_2$, this statistic follows a t-distribution with $df = n_1 + n_2 - 2$ degrees of freedom. The null hypothesis is rejected at significance level $\alpha = 0.05$ if the two-tailed p-value is less than α.

3.4 One-Way ANOVA and Kruskal-Wallis Test (RQ2)

When normality and homoscedasticity hold, one-way ANOVA (Fisher, 1925) is used to test for mean differences across the four parental education groups. The model is $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, where $\varepsilon_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$. The F-statistic is:

$$F = \frac{MS_{\text{Between}}}{MS_{\text{Within}}} = \frac{SS_{\text{Between}}/(k-1)}{SS_{\text{Within}}/(N-k)} \tag{6}$$

where $SS_{\text{Between}} = \sum_{i=1}^{k} n_i(\bar{x}_i - \bar{x})^2$ and $SS_{\text{Within}} = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2$, with $k = 4$ groups and N total observations. Under H_0, $F \sim F(k-1, N-k)$.

When normality is violated, the non-parametric Kruskal-Wallis test (Kruskal & Wallis, 1952) is applied instead. All N observations are ranked jointly, and the test statistic is:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(N+1) \tag{7}$$

where R_i is the rank sum of group i. Under H_0, $H \overset{\text{approx.}}{\sim} \chi^2(k-1)$ for $n_i \geq 5$.

3.5 Post-Hoc Analysis

A significant omnibus test (ANOVA or Kruskal-Wallis) indicates that at least one group differs from the others, but does not identify which pairs differ. Post-hoc tests are therefore required.

Tukey's Honestly Significant Difference (HSD) is applied after a significant ANOVA result (Tukey, 1949). For unequal group sizes, the Tukey-Kramer modification is used:

$$q = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{\dfrac{MS_{\text{Within}}}{2}\left(\dfrac{1}{n_i} + \dfrac{1}{n_j}\right)}} \tag{8}$$

This statistic q is compared against the critical value from the studentized range distribution $q_{\alpha,k,N-k}$. A pair (i, j) is declared significantly different if the corresponding p-value is below $\alpha = 0.05$.

Dunn's test with Bonferroni correction (Dunn, 1964) is applied after a significant Kruskal-Wallis result. For each pair of groups (i, j), the test statistic is:

$$z_{ij} = \frac{\bar{R}_i - \bar{R}_j}{SE_{ij}}, \qquad SE_{ij} = \sqrt{\frac{N(N+1)}{12}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)} \tag{9}$$

where $\bar{R}_i = R_i/n_i$ is the mean rank of group i. The raw p-value from the standard normal distribution is adjusted using the Bonferroni correction: $p_{\text{adjusted}} = \min(p_{\text{raw}} \times m,\ 1)$, where $m = \binom{k}{2}$ is the total number of pairwise comparisons. A pair is significant if $p_{\text{adjusted}} < 0.05$.

4 Evaluation

4.1 Descriptive Analysis

4.1.1 Overall Distribution and Scores by Gender

Overall, Mathematics scores range from 23 to 100 ($\bar{x} = 66.38$, $s = 15.05$, median = 66) and Language scores from 19.5 to 100 ($\bar{x} = 69.00$, $s = 14.45$, median = 70). When we split by gender (Table 1), a clear pattern appears: males outscore females in Mathematics by about five points on average ($\bar{x} = 68.95$ vs. 63.86), while females outscore males in Language by roughly eight points ($\bar{x} = 73.00$ vs. 64.93). The Language gap is noticeably wider. Standard deviations are similar across groups in both subjects, so these differences are not driven by one group being more spread out than the other.

Table 1: Descriptive statistics for Mathematics and Language scores by gender.

Subject	Gender	n	Mean	Median	SD	IQR
Mathematics	Female	245	63.86	64.00	15.26	20.00
	Male	241	68.95	70.00	14.43	22.00
Language	Female	245	73.00	73.00	14.01	18.50
	Male	241	64.93	65.50	13.75	21.00

The boxplots in Figure 1 tell the same story visually. There is substantial overlap — plenty of girls outscore plenty of boys in Mathematics and vice versa in Language — but the shift in the medians is hard to miss, especially for Language. The means (diamonds) sit close to the medians, which suggests the distributions are roughly symmetric rather than pulled by a few extreme scores.

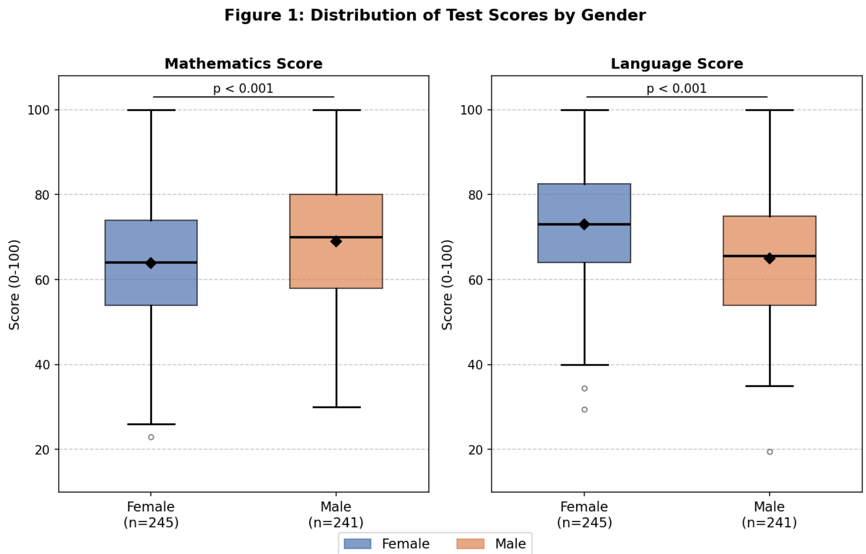

Figure 1: Boxplot of Mathematics and Language scores by gender. Diamonds indicate group means. Whiskers extend to $1.5 \times$ IQR; circles are outliers.

4.1.2 Scores by Parental Level of Education

Table 2 breaks the data down by parental education. Scores climb steadily from the high school group to the master's degree group in both subjects. What stands out, though, is that most of the jump happens between high school and associate's degree — after that, the gains are smaller. In Mathematics the mean goes from 62.24 (high school) to 67.81 (associate's), and then only creeps up to 69.75 (master's). In Language the pattern is similar but the overall spread is wider: 63.53 up to 75.53, a twelve-point gap. The master's degree group is also the smallest ($n = 59$), so its statistics carry a bit more uncertainty.

Table 2: Descriptive statistics for Mathematics and Language scores by parental level of education.

Subject	Parental Education	n	Mean	Median	SD	IQR
Mathematics	High School	156	62.24	63.00	14.04	17.25
	Associate's Deg.	177	67.81	66.00	15.24	24.00
	Bachelor's Deg.	94	68.44	67.00	15.05	16.75
	Master's Deg.	59	69.75	73.00	15.15	25.50
Language	High School	156	63.53	65.00	13.40	19.62
	Associate's Deg.	177	70.02	72.00	14.21	20.00
	Bachelor's Deg.	94	72.06	73.00	14.27	17.25
	Master's Deg.	59	75.53	76.00	13.55	19.50

Figure 2 makes this gradient easy to see. The high school box sits clearly below the rest in both panels, while the three higher-education groups overlap considerably with each other. Variability looks roughly similar across groups, which is a good sign for the equal-variance assumption we check in the next section.

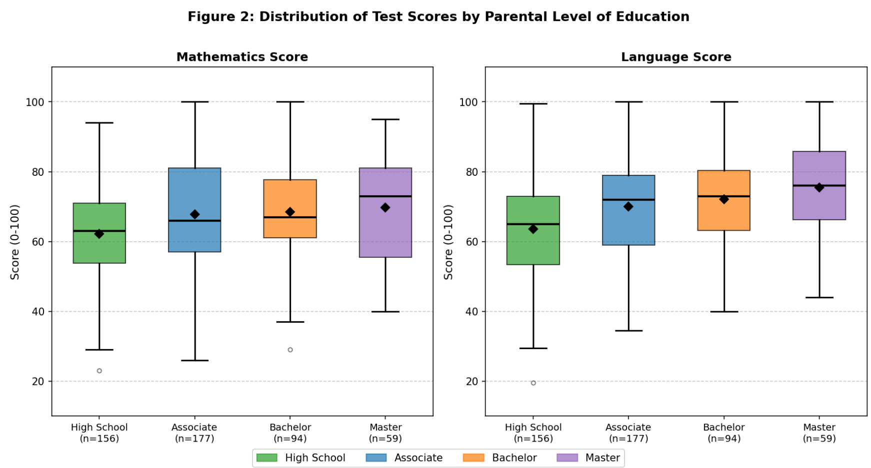

Figure 2: Boxplot of Mathematics and Language scores by parental level of education. Diamonds indicate group means. Parental education levels are ordered from lowest (High School) to highest (Master's Degree).

4.2 Inferential Analysis

4.2.1 Assumption Checks

Before choosing which hypothesis tests to run, we need to check whether the data meet the assumptions those tests require. Specifically: are the scores approximately normally distributed within each subgroup, and are the variances similar across groups?

For the gender comparison (RQ1), the Shapiro-Wilk test does not reject normality in any of the four subgroups: female Mathematics ($W = 0.9944$, $p = 0.501$), male Mathematics ($W = 0.9885$, $p = 0.050$), female Language ($W = 0.9899$, $p = 0.085$), and male Language ($W = 0.9909$, $p = 0.139$). The male Mathematics result deserves a note — $p = 0.050$ is right on the boundary, so normality is technically not rejected, but only just. Levene's test shows no variance problems at all: $F = 0.006$, $p = 0.938$ for Mathematics and $F = 0.160$, $p = 0.690$ for Language. With both assumptions satisfied, the two-sample t-test is appropriate for RQ1.

For the parental education comparison (RQ2), the picture is mixed. Two of the four Mathematics subgroups fail the normality check: the associate's degree group ($W = 0.9846$, $p = 0.049$) and the master's degree group ($W = 0.9559$, $p = 0.032$). Because a parametric ANOVA assumes all groups are normal, we use the non-parametric Kruskal-Wallis test for Mathematics instead. Language scores are better behaved — all four subgroups pass normality, and Levene's test confirms equal variances ($F = 0.169$, $p = 0.917$) — so the one-way ANOVA can be used there.

4.2.2 RQ1: Gender Differences

We test whether the mean scores differ between females and males, separately for each subject. The hypotheses are the same in both cases:

- H_0: $\mu_{\text{female}} = \mu_{\text{male}}$
- H_1: $\mu_{\text{female}} \neq \mu_{\text{male}}$

For Mathematics, $t(484) = -3.78$ with $p = 0.000179$. We reject H_0: males score significantly higher ($\bar{x} = 68.95$ vs. 63.86). Cohen's $d = -0.34$, a small-to-medium effect (Cohen, 1988). The gap is real but not enormous — roughly a third of a standard deviation.

For Language the result is even clearer: $t(484) = 6.41$, $p < 0.0001$. Females score significantly higher ($\bar{x} = 73.00$ vs. 64.93), and the effect size $d = 0.58$ is medium-to-large — almost twice the magnitude of the Mathematics gap. So while both differences are statistically significant, the Language gap is the more practically meaningful one.

4.2.3 RQ2: Parental Education Differences

Mathematics scores. Because two subgroups failed the normality check, we use the Kruskal-Wallis test here:

- H_0: The distribution of Mathematics scores is identical across all four parental education groups.
- H_1: At least one group differs.

The result is significant: $H = 15.68$, $df = 3$, $p = 0.0013$. Scores are not the same across education levels. But which groups actually differ? Dunn's test with Bonferroni correction across all $\binom{4}{2} = 6$ pairs (Table 3, adjusted p-values compared against $\alpha = 0.05$) shows that the high school group is the outlier — it differs significantly from all three higher-education groups. The associate's, bachelor's, and master's groups, by contrast, are statistically indistinguishable from each other.

Table 3: Post-hoc pairwise comparisons for Mathematics scores (Dunn's test, Bonferroni correction).

Group 1	Group 2	z	p_{raw}	p_{adj}	Significant
High School	Associate's Deg.	-3.033	0.0024	0.0145	Yes
High School	Bachelor's Deg.	-2.972	0.0030	0.0178	Yes
High School	Master's Deg.	-3.091	0.0020	0.0120	Yes
Associate's Deg.	Bachelor's Deg.	-0.431	0.6668	1.000	No
Associate's Deg.	Master's Deg.	-0.927	0.3538	1.000	No
Bachelor's Deg.	Master's Deg.	-0.508	0.6112	1.000	No

Language scores. All four subgroups passed normality and Levene's test, so we can use the one-way ANOVA:

- H_0: $\mu_{\text{HS}} = \mu_{\text{AD}} = \mu_{\text{BD}} = \mu_{\text{MD}}$
- H_1: At least one group mean differs.

The ANOVA is highly significant: $F(3,482) = 14.24$, $p < 0.0001$. Tukey's HSD (Table 4) reveals a pattern that is largely similar to Mathematics but with one extra finding.

Table 4: Post-hoc pairwise comparisons for Language scores (Tukey's HSD). $MS_{\text{Within}} = 192.90$, $df_{\text{within}} = 482$.

Group 1	Group 2	Mean diff.	p-value	Significant
High School	Associate's Deg.	-6.49	0.0001	Yes
High School	Bachelor's Deg.	-8.53	< 0.0001	Yes
High School	Master's Deg.	-11.99	< 0.0001	Yes
Associate's Deg.	Bachelor's Deg.	-2.04	0.6596	No
Associate's Deg.	Master's Deg.	-5.50	0.0429	Yes
Bachelor's Deg.	Master's Deg.	-3.47	0.4365	No

Again, the high school group scores significantly lower than all three higher-education groups. But unlike in Mathematics, there is also a significant gap between the associate's and master's degree groups ($p = 0.0429$). The bachelor's group sits in between and does not differ significantly from either neighbour. This suggests that for Language, the parental education gradient extends a bit further up the scale than it does for Mathematics.

5 Summary and Discussion

We set out to answer two questions: whether boys and girls perform differently in Mathematics and Language, and whether parental education matters. The data say yes to both.

On gender (RQ1), males score about five points higher in Mathematics ($t(484) = -3.78$, $p < 0.001$, $d = -0.34$) while females score about eight points higher in Language ($t(484) = 6.41$, $p < 0.001$, $d = 0.58$). The Language gap is the larger of the two in practical terms. These results are consistent with what PISA reports internationally (OECD, 2023), though the size of the gap varies from one context to another.

On parental education (RQ2), the story is clear: the high school group falls behind in both subjects. In Mathematics ($H = 15.68$, $p = 0.0013$), Dunn's test shows the high school group is significantly lower than all three higher-education groups, which themselves are statistically indistinguishable. In Language ($F(3,482) = 14.24$, $p < 0.0001$), Tukey's HSD tells a similar story but adds one extra finding — the associate's and master's degree groups also differ from each other. The biggest takeaway is that the critical divide is between having no post-secondary education and having any at all, rather than between progressively higher degrees. This fits the broader literature on socioeconomic gradients in education (Sirin, 2005).

There are things this analysis cannot tell us. We have no data on income, school quality, or how much time students spent studying — any of which could confound the patterns we see. The male Mathematics normality result ($p = 0.050$) was right on the edge, so the t-test conclusion there, while technically justified, should not be over-interpreted. And because the data are cross-sectional, we cannot say that parental education *causes* higher scores — only that the two are associated.

Looking ahead, three directions seem worthwhile:

1. **Targeted support for first-generation students.** The high school parent group consistently scores lowest. Tutoring or mentoring programmes aimed specifically at these students could help close the gap before it widens further.
2. **Digging into the gender gap.** Boys do better in Mathematics, girls do better in Language — but why? Investigating whether confidence, study habits, or teaching methods play a role would turn a statistical pattern into something schools can act on.
3. **Policy implications.** If the key threshold is between no tertiary education and any tertiary education, then policies that help more parents access higher education could have knock-on benefits for their children — an intergenerational payoff worth exploring.

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7 Appendix

Code: `https://github.com/hirarehman161/Gender-and-Parental-Education-Effects-on-Student-Test-Performance`